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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

IS2201-1- INFORMATION STORAGE MANAGEMENT

(For IS) .

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. In a cloud computing environment, which service model provides complete infrastructure services such as storage and networking?
A) SaaS
B) PaaS
C) IaaS
D) DaaS
2. Which storage technology enables multiple physical storage devices to appear as a single storage unit?
A) Direct Attached Storage
B) Storage Virtualization
C) Optical Storage
D) Magnetic Storage
3. Which of the following is NOT a key benefit of RAID implementation?
A) Improved performance
B) Fault tolerance
C) Data encryption
D) Increased storage efficiency
4. Which RAID level uses striping without redundancy?
A) RAID 0
B) RAID 1
C) RAID 5
D) RAID 6
5. Which of the following RAID levels offers the best balance between redundancy and storage efficiency?
A) RAID 0
B) RAID 1
C) RAID 5
D) RAID 6
6. What is the purpose of snapshots in intelligent storage systems?
A) Improve data security
B) Provide point-in-time data copies
C) Enhance read/write speeds
D) Compress stored data
7. Which of the following is an example of Direct-Attached Storage (DAS)?
A) Network-Attached Storage (NAS)
B) Storage Area Network (SAN)
C) Internal hard drive
D) Cloud storage
8. What is one key benefit of Parallel SCSI over earlier storage interfaces?
A) Lower cost
B) Higher data transfer rates
C) Simpler cabling
D) Wireless connectivity
9. What does FC in FC SAN stand for?
A) Fibre Connector
B) Fibre Channel
C) Fast Channel
D) File Connection
10. Which of the following is an FC SAN topology?
A) Star
B) Bus
C) Switched Fabric
D) Ring

11. Which layer of the Fibre Channel protocol is responsible for mapping upper-layer protocols?
 - A) FC-0
 - B) FC-1
 - C) FC-2
 - D) FC-4
12. Which protocol is commonly used for NAS file sharing in Windows environments?
 - A) NFS
 - B) SMB
 - C) iSCSI
 - D) FCIP
13. What is a major factor affecting NAS performance?
 - A) Storage capacity
 - B) Network bandwidth
 - C) CPU speed
 - D) Power supply
14. Which protocol is commonly used for accessing Object-Based Storage?
 - A) NFS
 - B) SMB
 - C) REST API
 - D) SCSI
15. What type of port in an FC SAN is used to connect to another switch?
 - A) F_Port
 - B) N_Port
 - C) E_Port
 - D) L_Port
16. Which iSCSI component serves as the storage provider?
 - A) iSCSI Initiator
 - B) iSCSI Target
 - C) iSCSI Switch
 - D) iSCSI Gateway
17. Which backup strategy minimizes the amount of storage space required?
 - A) Full backup only
 - B) Incremental backup
 - C) Continuous data protection (CDP)
 - D) Snapshot backup
18. What does a disaster recovery plan primarily focus on?
 - A) Preventing cyber attacks
 - B) Ensuring long-term data archiving
 - C) Restoring IT operations after a disruption
 - D) Increasing storage efficiency
19. Which cloud deployment model is dedicated to a single organization?
 - A) Public cloud
 - B) Private cloud
 - C) Hybrid cloud
 - D) Community cloud
20. Which of the following is a common method used for securing data at rest in storage networks?
 - A) Data replication
 - B) RAID 0
 - C) Encryption
 - D) Compression

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe the key components of a data center infrastructure and explain their roles	8	L*2	1	1,3
	b) Analyze the impact of virtualization on storage architecture. How does it improve efficiency and resource utilization?	8	L2	1	1,3
2.	a) Explain RAID levels 0 and 1 with diagrams and use cases.	8	L2	2	1,3
	b) A company wants to implement RAID for high availability and performance. Which RAID level would you recommend and why?	8	L3	2	1,3
3.	a) Explain the role of cache memory in an Intelligent Storage System and its impact on performance	8	L2	2	1,3
	b) Explain the types of Direct-Attached Storage (DAS) with examples.	8	L2	2	1,3
Unit – II					
4.	a) Describe the key components of an FC SAN and their roles in data storage.	8	L2	3	1,2
	b) Analyze the importance of Zoning in FC SANs. How does it improve security and traffic management?	8	L3	3	1,2
5.	a) Describe the components of NAS and their functions.	8	L2	3	2,3

b)	Explain the working of iSCSI in IP SANs and its advantages over Fibre Channel.	8	L2	4	1,2
6. a)	Explain the concept of Object-Based Storage (OBS) and how it differs from traditional file and block storage.	8	L2	4	1,2
b)	Analyze the role of Unified Storage in modern IT infrastructure. How does it improve efficiency and resource utilization?	8	L3	4	2,3
Unit – III					
7. a)	What is Failure Analysis, and how does it help in designing a resilient IT infrastructure?	8	L2	5	1,2
b)	What are backup targets? Compare different backup storage options such as tape, disk, and cloud.	8	L3	5	1,2
8. a)	Explain the key cloud-enabling technologies and how they support cloud computing.	8	L2	5	1,2
b)	Explain the importance of data integrity mechanisms in storage security. How can organizations ensure data remains untampered?	8	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
